ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B.TECH (CSE) (AI)

Semester: IV

SUBJECT CODE: BCSAI403

DATABASE MANAGEMENT SYSTEM

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. (10 Marks)

Q. 1. Answer the following questions: **(1 × 5 = 5 Marks)**

(i) Explain about deadlocks.
(ii) Explain about different DCL operations.
(iii) Explain the role of a data base administrator.
(iv) What are the different levels of abstraction?
(v) Explain the terms super key and candidate key with an example.

Q. 2. Answer the following questions: **(1 × 5 = 5 Marks)**

(i) What is a table in SQL?
(ii) Explain the purpose of a trigger in SQL.
(iii) What is a cursor in SQL and when is it used?
(iv) What is the difference between the COUNT and SUM aggregate functions?
(v) Explain the Two-Phase Locking (2PL) protocol in concurrency control.

SECTION – B

NOTE:Attempt all questions. **(5×2×2 = 20 Marks)**

Q.3.(i)(a) Define a Database Management System and its key features. What is the difference between a Database System and a File System? Provide at least three points of comparison.

OR

(b) Describe the architecture of a database system. Explain the concepts of schema and instance in a DBMS. How are they different from each other?

(ii)(a) Explain the importance of data models in a DBMS. Provide examples of different types of data models.

OR

(b) What are the common symbols used to represent entities, relationships, and attributes? Discuss the different types of mapping constraints in an ER diagram. Provide examples of each type.

Q.4. (i)(a) Write SQL queries based on a relational database schema to demonstrate the use of DML operations such as SELECT, INSERT, UPDATE, and DELETE.

OR

(b) Design an ER diagram for a library management system, and map the design into a relational schema. Include entities such as books, authors, borrowers, and transactions.

(ii)(a) Describe the types of joins in SQL. Explain INNER JOIN, LEFT JOIN, RIGHT JOIN, and FULL JOIN with examples.

OR

(b) What are SQL operators? Describe the function of arithmetic, comparison, and logical operators with examples.

SECTION – C

NOTE: Attempt all questions. **(10 × 2 = 20Marks)**

Q.5. (a). Discuss the process of converting an ER diagram to a relational schema. Explain the concept of a Relationship of Higher Degree. How can this be represented in the ER model?

OR

(b) Explain the concept of normalization in database design. What is a lossless join decomposition in database normalization? Explain why it is important for preserving data integrity when decomposing relations.

Q.6. (a) What is a transaction in the context of a Database Management System (DBMS)? Explain the ACID properties of transactions and their significance.

OR

(b) What is concurrency control in database systems? Why is it necessary for maintaining data consistency and isolation in a multi-user environment?